

LEDGER · 2026-09-04 · MATERIALS CHEMISTRY

Ultrafast Periodate-Accelerated Polycatecholamine Aqueous Adhesion Promoters via Oxidative Crosslinking

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	3M Tape Primer 94
Entry application	wet-environment structural adhesion and VOC-restricted automotive/architectural detailing
Measured advantage	none
Time to first revenue	12 months
Gross margin at parity	94.7%
Startup CapEx	\$255,000
Payback from first sale	0.1 months
Capital productivity	223.15x

Why it failed

- Voided by: Economics undisclosed, Negative control / sub-2x, Entry application mismatch, Form-factor / handling regression, Absence search missing
- Absence asserted, search unreported

Assessment

The report explicitly concedes failure on superiority and CapEx. Startup CapEx of \$255,000 exceeds the \$250,000 threshold, the product suffers form factor regression requiring a two-part system, thermal breakdown above 50°C undermines application fitness, and the market absence ledger is missing.

Also flagged

- Duplicate references inflating the evidence base
- Price parity asserted, not sourced
- runsheet_no_quantities
- form_factor_regression

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-09-04 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

safety controls. Independently verify every figure, citation and regulatory requirement before committing capital.